

Docker Interview Questions & Answers

Simple Hinglish • Placement Focus • Unnecessary Advanced Topics Removed

Core idea: Docker ko 4 questions se samjho — Image kya hai? Kaunsa process run ho raha hai? Network me kisse baat karega? Host se kaunsa port publish hoga?

1. Docker kya hai?

Docker ek containerization platform hai jo application ko uski dependencies ke saath isolated environment me run karne deta hai.

Interview: **What is Docker?**

Docker is a containerization platform used to package and run applications in isolated containers along with their dependencies.

2. Docker Image kya hai?

Image = blueprint/template. Isse containers create hote hain. Image generally read-only hoti hai.

Interview: **What is a Docker Image?**

A Docker image is a read-only blueprint/template used to create Docker containers.

3. Docker Container kya hai?

Container = image ka running instance. Ek image se multiple containers ban sakte hain.

Interview: **Image vs Container?**

Image is a blueprint, while a container is a runnable/running instance created from that image.

4. Dockerfile kya hai?

Dockerfile image banane ki instructions wali text file hai.

Interview: **What is a Dockerfile?**

Dockerfile is a text file containing instructions used to build a Docker image.

5. Docker Daemon kya hai?

Docker Daemon (dockerd) background service hai jo containers, images, networks etc. manage karta hai.

Interview: **What is Docker Daemon?**

Docker Daemon is a persistent background process that manages Docker containers and related resources.

6. Docker Engine kya hai?

Docker Engine Docker ka core system hai jo images build karne aur containers run/manage karne ka kaam karta hai.

Interview: **What is Docker Engine?**

Docker Engine is the core technology used to build, run and manage containers.

7. Docker CLI kya hai?

CLI se hum Docker commands chalte hain, jaise docker ps, docker run, docker stop.

Interview: **What is Docker CLI?**

Docker CLI is the command-line interface used to interact with Docker.

8. Docker Image Layers kya hain?

Image multiple layers se bani hoti hai. Layers reusable/cached hoti hain, isliye rebuild faster ho sakta hai.

Interview: **What are Docker image layers?**

Docker images consist of multiple filesystem layers. Layers can be reused through build cache.

9. docker run kya karta hai?

docker run new container create karke start karta hai.

Command: **docker run nginx**

Interview: It creates and starts a new container from an image.

10. -d kya hai?

-d = detached mode. Container background me run hota hai.

Command: **docker run -d nginx**

Interview: Detached mode runs a container in the background.

11. -it kya hai?

-i input open rakhta hai, -t terminal allocate karta hai. -it interactive terminal ke liye useful hai.

Command: **docker run -it ubuntu bash**

Interview: -it provides an interactive terminal.

12. Port Mapping kya hai?

Syntax: **-p HOST_PORT:CONTAINER_PORT**. Example: -p 8080:80 means host 8080 → container 80.

Interview: **Explain -p 8080:80.**

It maps host port 8080 to container port 80.

13. EXPOSE vs -p

EXPOSE image me expected listening port ko document karta hai; -p actual host-to-container port publishing karta hai.

Interview: **Does EXPOSE publish a port?**

No. EXPOSE documents the port; -p publishes/maps it to the host.

14. Environment Variables / -e

Runtime configuration dene ke liye -e use hota hai. Example: -e PORT=5000.

Interview: **Why use -e?**

It passes environment variables to a container at runtime.

15. Docker Network kya hai?

Docker network containers ko controlled way me aapas me communicate karne deta hai.

Interview: **Why use Docker networks?**

They allow containers to communicate with each other in an isolated and controlled environment.

16. Bridge Network

Local multi-container applications ke liye user-defined bridge network commonly useful hai. Same network par container names se communication possible hota hai.

Interview: **What is a bridge network?**

It provides networking between containers, commonly for local multi-container applications.

17. localhost Docker me

Container ke andar localhost ka matlab wahi container hota hai. Dusre container ke liye shared network par uska hostname/container name use karo.

Interview: **Why not localhost for another container?**

Because localhost inside a container refers to that same container.

18. Docker Volume

Volume container ke bahar persistent data rakhne ke liye use hota hai. Database data ke liye important hai.

Interview: **Why use Docker volumes?**

To persist data outside the container lifecycle.

19. Container Exited kyun hota hai?

Container ka lifecycle main process/PID 1 se tied hota hai. Main process exit karega to container generally stop ho jayega.

Common codes: **0** = successful completion, **1** = application/runtime error, **137** = often SIGKILL/memory kill.

20. docker ps vs docker ps -a

docker ps = running containers. **docker ps -a** = all containers, including stopped/exited.

Interview: docker ps shows running containers; docker ps -a shows all containers.

21. docker logs

Container ki application output/errors dekhne ke liye use hota hai.

Commands: **docker logs NAME** and **docker logs -f NAME** for live logs.

22. docker exec

Running container ke andar command/shell chalane ke liye.

Command: **docker exec -it NAME sh**

Important: generally only running containers.

23. docker inspect

Container ki detailed configuration, environment, network, ports, mounts etc. dekhne ke liye.

Command: **docker inspect NAME**

24. Docker Hub

Docker Hub ek public container registry hai jahan se images pull ki ja sakti hain.

Command: **docker pull nginx**

Interview: Docker Hub is a cloud-based registry for storing and downloading container images.

25. docker pull vs docker run

docker pull image download karta hai. **docker run** image se container create + start karta hai.

Example: docker pull nginx → image local me; docker run nginx → container start.

Important Docker Commands — Must Remember

Command	Simple Meaning
docker images	Local images list
docker ps	Running containers
docker ps -a	All containers
docker pull IMAGE	Image download
docker run IMAGE	New container create + start
docker start NAME	Stopped container start
docker stop NAME	Container stop
docker restart NAME	Restart container
docker rm NAME	Container delete
docker rmi IMAGE	Image delete
docker logs NAME	Container logs
docker exec -it NAME sh	Running container ke andar shell
docker inspect NAME	Detailed information
docker stats	CPU/RAM/resource usage
docker port NAME	Port mapping check
docker network ls	Networks list
docker network create NET	Network create
docker network inspect NET	Network details
docker system df	Docker disk usage
docker system prune	Unused resources cleanup — carefully

Top 15 Questions for Placement

1. What is Docker?
2. What is a Docker Image?
3. What is a Docker Container?
4. Image vs Container?
5. What is Dockerfile?
6. What is Docker Daemon?
7. What is Docker Engine?
8. What does docker run do?
9. Difference between -it and -d?
10. Explain -p 8080:80.
11. EXPOSE vs -p?
12. Why use environment variables / -e?
13. What is Docker Network?
14. Why does localhost not connect to another container?

15. Why does a container show Exited?

Topics intentionally kept low priority

macvlan, ipvlan, deep overlay networking, docker events, docker diff, docker top, advanced BuildKit, advanced registry authentication, complex Compose settings, and other deep internals are not included in the main interview list. For basic placement interviews, focus first on images, containers, Dockerfile, daemon/engine, run modes, ports, environment variables, networks, volumes, logs, exec, inspect and troubleshooting.

Quick Memory Map

Dockerfile → **Image** → **docker run** → **Container** → **Process**

-d = background

-it = interactive terminal

-e = environment variable

-p HOST:CONTAINER = port mapping

Network = container-to-container communication

logs = error/output check

exec = container ke andar command

inspect = detailed configuration